

Chapter 4: Testing

4.1 Testing Strategy

A structured testing process was established to provide sufficient evidence of the reliability of the brain tumor classification system for real-world deployment. The approach taken to testing was to break down the evaluation process into separate stages instead of testing all parts of the system simultaneously. Initial evaluations were conducted for the enhancement module alone; followed by an evaluation for each classification model individually; and finally, an evaluation was conducted for the entire pipeline in an end-to-end fashion. This systematic manner of completing evaluations provided opportunities to detect issues in the early stages, ensure that all system components operated as intended, and confirm that the integrated pipeline produced consistent results across varying environments.

4.1.1 Testing Framework

The evaluation's structure consisted of three phases to ensure that each portion of the system underwent sufficient inspection prior to assembling the complete system.

Phase 1 - Testing Enhancement Models

We evaluated three different enhancement networks, U-Net, DnCNN, and EDSR. The three models were trained using the exact same training and validation data so that the results from this phase could be compared evenly, and the test data was only accessed during the final phase of the evaluation. Performance was assessed using standard image quality metrics: Peak Signal to Noise Ratio (PSNR) and Structural Similarity Index (SSIM), calculated using the widely accepted scikit-image library. This phase was extremely important because a noisy or poorly enhanced image may mislead the classifier at a later time. We paid close attention to how effectively each network model cleaned up artifacts while not blurring or distorting clinically relevant anatomical information.

Phase 2 – Testing the Classification Models

We trained and evaluated five classification architectures after selecting the best-performing enhancement model (U-Net in our case). All classifiers were therefore trained on the same high-quality, consistent enhanced images. In addition to reporting total accuracy, we provided a comprehensive set of clinically relevant metrics, such as precision, recall, F1 (macro and weighted), specificity, and AUC-ROC (using scikit-learn) which provided clear insights into how each model performed with respect to the rarer tumor types, which is critical in a medical context where missing any one of these can have dire consequences.

Phase 3 – End-to-End Integration and Deployment Testing

During the last phase of our work we integrated our best-performing image enhancement and classification models into a single pipeline and created a Gradio web interface to run the pipeline. We then evaluated the entire workflow beginning with image upload, following through preprocessing and enhancement and concluding with predicting and displaying the prediction result from many different sources of real-world images (different types of files, file sizes, variations in noise and artifacts from scanning). This allowed us to validate whether or not the solution is robustly usable by others outside of our lab once it is deployed and operated.

Our phased approach allowed us to identify and address problems very early on, instilled confidence in the performance of each individual component in isolation, and ultimately produced a system capable of functioning reliably throughout the entire workflow.

4.1.2 Tools and Technologies

We used various tools to simplify and increase the reliability of the testing process:

- TensorFlow/Keras was the core platform used to create, train, and test deep learning models. The built-in metrics, auto-checkpointing (ModelCheckpoint callbacks), and mixed-precision support enabled us to be less error-prone and focus on other things; we didn't need to constantly remember to save the best weights.
- Scikit-learn provided a way to evaluate the results, using its `'classification_report'`, `'confusion_matrix'`, and `'roc_auc_score'` functions, which generate comparable, exact numbers for all other researchers publishing similar results.
- Matplotlib/Seaborn provided visual visualizations, generating more detailed confusion matrix, ROC curves, and side-by-side comparison charts. This allowed us to quickly pinpoint model weaknesses, such as which tumor types/models were being confused the most frequently.
- Google Colaboratory was the primary environment we will be using to develop our code, because it allowed us to use free access to both GPU/TPU resources so we could try out many types of model architectures and hyper-parameters without limit based upon the specifications of our hardware at home. We also created our own Google Drive folder for the model we were developing which allowed us to easily link all of our model logs, trained models and post-analysis plots together with ease of access.

By using these integrated tools, we were able to create a well-sorted, repeatable testing period that provided an efficient way to test all through the development of the models we would create.

4.1.3 Data Splitting Strategy

To achieve accurate results, it was important that we had the correct split of data to use for training our model(s) and testing them. We utilized a very standard method to ensure a clean data split. The following were the data splits used in descending order:

- The training set is composed of 70% of all the data available and will be used only to allow the model(s) to run algorithms through the training process to learn the patterns that exist.
- The validation set is composed of 20% of data that will be monitored during the training process for tuning hyperparameters, scheduling learning rates, and determining the early-stopping point of the training process.
- The test set is composed of the remaining 10% of data that was taken from the original dataset and was not used in any way before the last evaluation of the model(s).

The separation of the test set from the original data was completed before any data preprocessing or model testing was completed to minimize the risk of creating a "leak" to the training set, which could inflate the results reported on the test set.

When the dataset allowed, we also ensured that the patients were separated at the level of each patient: If there were multiple scans from a single patient, they all went into a single split rather than being separated between training and testing datasets. By this means, our model(s) could not find certain characteristics of the individual patient to memorise or identify, thus ensuring that they learned to identify generalisable characteristics of all the patients. By following such a stricter separation of the different datasets, the overall results that we achieved were much more representative of what would be clinically applicable.

4.1.4 Testing Protocols

The models were not just trained and left to chance. Instead, each model architecture underwent a thorough evaluation process consisting of a series of evaluations, both quantitative and qualitative. Quantitative Evaluation consisted of recording and analyzing all the significant numerical values that were tracked throughout the training process, including the loss and accuracy of the training and validation datasets as well as the final test metrics associated with each model. In addition, all of this data was recorded into CSV files so that

we could go back and analyze it later and create a visual representation of historical trends, and also allow for fair comparisons between various models.

The qualitative evaluation included manually examining the predicted outputs of each model using a random sample from the test dataset. It became evident very quickly if a model had truly captured the anatomical structures of interest, or if it was using other means (such as horse detection training programs or background images) to obtain the desired predictions. To properly assess each model's performance, we artificially introduced many real-world hurdles through for example scans filled with noise, slice orientations that are not typical, small tumour sizes, large tumours that significantly distort normal anatomy, and/or scans containing metal artifacts. By testing these models under these extreme scenarios, we identified the actual limits of each structure within the real world and found out exactly which aspects of the system will require human oversight before it can be used.

To ensure that results were able to be consistently replicated, we performed reproducibility checks using a fixed random seed (i.e. 42) across each experiment, including data shuffling, augmentation and weight initialisation processes. We confirmed that any time an identical notebook was run, we received perfectly replicated results (i.e. memorised zeroes). This level of determinism allows anyone to independently verify or further develop our research, thereby making the entire investigation reproducible.

Evaluating each model based on these four separate avenues provided us with a clearer, more authentic representation of how the model will likely perform when it's released to the public.

4.2 Test Cases and Outcomes

4.2.1 Enhancement Models Test Cases

Test Case 1: Standard PSNR and SSIM Evaluation

The goal was to compare three different enhancement models by measuring how well the models enhanced image quality while preserving structure. Every enhancement model received the same set of 40 different types of noisy test images for analysis. In doing so, we measured Peak Signal-to-noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) by comparing each model's enhanced output and its corresponding clean reference (ground-truth) image. This experiment provided an unbiased and objective measure of the

model's ability to remove noise from test images while preserving any important anatomical features of the original images.

Results:

Model	PSNR (dB)	SSIM
U-Net	32.453	0.9066
DnCNN	31.701	0.8919
EDSR	32.469	0.8900

Table 4.1: Performance comparison Of enhancement models on the test set.

The U-Net was determined to be the best model based on the highest SSIM (Structure Similarity Index) score, which means it produced the most accurate representation of the anatomical structure of the brain; this is essential for making most medical imaging decisions. While EDSR slightly outperformed U-Net in PSNR (Peak Signal to Noise Ratio), the difference was small, and likely would not have affected actual clinical use. Since the structural representation is of far greater value than the amount of noise present at the pixel level when using a neural augmentation model to enhance medical image quality, we chose U-Net for the improved image enhancement portion of this project.

The qualitative view of images produced further confirmed the findings from the quantitative results. U-Net images were sharp and natural; there were clearly defined tissue boundaries, and even small-scale anatomical features were visible. When using DnCNN, we lost the integrity of the texture of the images because the model over-smoothed everything. EDSR produced clean images overall but occasionally had faint ring-like artifacts near edges of the images with high contrast. These artifacts were not severe but were present enough that we did not consider EDSR for clinical applications.

Test Case 2: Computational Efficiency

This research project's objective was to calculate the time needed for one pass of each of the three enhancement models when applied to an average-sized image (224 x 224 pixels) and examine the usefulness of each model in commercial applications.

To determine the inference times of each model, we measured inference times for 100 passes using Tesla T4 GPUs hosted on Google Colab for groups of 8 images.

Findings were as follows:

- U-Net - Approximately 12.36 seconds to complete all 100 passes for Group A
- DnCNN - Approximately 12.53 seconds to complete all 100 passes for Group A
- EDSR - Approximately 13.50 seconds to complete all 100 passes for Group A

All three models had very similar inference times when used to enhance along with U-Net producing images of greater quality without added processing delays.

Test Case 3: Enhancement Impact on Classification

Objective: To examine whether the utilized enhancement modification is merely aesthetic in nature or does it actually provide a benefit to improved classification accuracy.

Methodology: For this study we have developed an original raw image based baseline CNN classifier twice under identical conditions using the same hyperparameters, splitting approach and random seed when comparing results to the modified U-Net images.

Results:

- Accuracy obtained from raw image inputs was 86.3%
- Accuracy obtained from U-Net-enhanced images was 89.5%

Conclusion: There was a statistical difference of 3.2% between the original raw image input and the U-Net modified images indicating that the enhancement was not only cosmetic in nature but that utilising cleaner, better structured images improves the accuracy of the classification of tumours and therefore supports the integration of an enhancement component into the final classification pipeline.

4.2.2 Classification Models Test Cases

Test Case 4: Multi-Model Classification Comparison

The goal of this study was to identify the best classification architecture for detecting brain tumors. The five models (The CNN, ResNet50, VGG16, Xception, EfficientNetB4) were implemented using identical training methodology including using UNet augmentation images with the same set of hyperparameters, Data Augmentation techniques and time frames for training. Each of the five Models were evaluated using the same held out test sets.

Results:

Model	Accuracy	Precision	Recall	F1-Score
CNN (Baseline)	89.49%	89.60%	89.49%	89.54%
ResNet50	96.17%	96.23%	96.17%	96.18%
VGG16	91.13%	92.21%	91.13%	91.21%

Xception	96.17%	96.23%	96.17%	96.18%
EfficientNetB4	98.17%	98.10%	98.03%	98.05%

Table 4.2: Global performance metrics of classification models on the test set.

The EfficientNetB4 was the strongest-performing model evaluated in this research project. On the held-out test set comprising 1,311 images, the EfficientNetB4 achieved accuracy of 97.94 %. This represents nearly a 2% increase compared to both ResNet50 and Xception, whose accuracies were 96.17 % each, as measured by the same test set. The statistical significance associated with the accuracies computed from the test set indicates that finding an additional 26 misclassifications per sample is meaningful in a clinical setting, where even modest improvements could have major effects on the health of patients.

Key Observations

- The baseline CNN achieved 100 percent accuracy during training but only 89.49 percent during testing, indicating clear overfitting. Its overall size and complexity were likely not sufficient for generalization across unseen training samples.
- Both ResNet50 and Xception produced roughly comparable results, achieving an accuracy of roughly 96.17 percent on test data, which was good but would, without doubt, enable improvement.
- At lower performance, VGG16 performed noticeably less than the other two networks due to the fact that its deeper structure, narrower node usage across layers, and large total number of model parameters will require substantially more training data than was provided.
- The combination of EfficientNetB4's compound scaling (e.g., balanced scaling of all three types of features, namely width, depth, and resolution) toward the appropriate size and complexity of a single dataset produced a model that is capable of extracting far more discriminating features than candidates requiring an unreasonable quantity of training data and compute resources.

For all these reasons, we selected an EfficientNetB4-based general model due to the well-established success of having higher accuracy and lower error rates due to fewer samples necessary per class, coupled with lower inference times and hence much improved speed for model deployment using U-Net-enhanced preprocessing .

Test Case 5: Per-Class Performance Analysis

Goal: Make sure that the model performs uniformly through all four tumor categories.

Approach: We calculated class-specific performance metrics of our best model (EfficientNetB4) to identify any systematic bias within our results.

Results:

Class	Precision	Recall	F1-Score	Specificity	Support
Glioma	94.67%	94.67%	94.67%	99.80%	300
Meningioma	95.24%	98.04%	96.61%	98.51%	306
No Tumor	99.75%	99.75%	99.75%	99.78%	405
Pituitary	98.36%	99.67%	99.01%	99.51%	300

Table 4.3: Per-Class Performance Analysis of EfficientNetB4 Model

The model performed exceptionally well in all classes. The "No Tumor" class attained almost perfect scores, which is unsurprising since it has a highly distinct visual appearance. In contrast, Gliomas performed somewhat less successfully as would be expected based on previous medical literature regarding the diverse appearances of these tumors. Most importantly, all specificities were $> 98\%$, indicating rarely producing false positives within any of the classes.

Test Case 6: Confusion Matrix Analysis

Aim: To discover the confusion when identifying different types of tumors. Methods: The confusion matrix was created from the predictions made by EfficientNetB4 on the test dataset and analysed for off-diagonal values.

Results:

	Pred: Glioma	Pred: Meningioma	Pred: No Tumor	Pred: Pituitary
True: Glioma	284	14	0	2

True: Meningioma	2	300	2	2
True: No Tumor	0	0	404	1
True: Pituitary	0	1	0	299

Table 4.4: Confusion Matrix Analysis of EfficientNetB4 Model

The results of the confusion matrix show interesting patterns. Gliomas and meningiomas are often confused (14 instances), which makes sense due to the similarities between both types of brain tumours. Of utmost importance, there were no instances of misclassification of non-tumour samples as tumour samples (only 1 false positive), which is the most severe type of misclassification to occur with respect to medical screening.

Test Case 7: ROC Curve and AUC Analysis

The purpose of this evaluation was to test the model's ability to distinguish between different types of tumors at various point estimates of probability. To accomplish this, we generated Receiver Operating Characteristic (ROC) curves for all classes (tumor types) using the one-vs-the-rest method and then computed the AUC for each of those curves.

Results:

Class	AUC
Glioma	0.9993
Meningioma	0.9981
No Tumor	0.9999
Pituitary	0.9999
Macro Average	0.9993

Table 4.5: ROC Curve and AUC Values of EfficientNetB4 Model

The results show that the computed AUCs were all greater than or equal to 0.99, which means that the model could differentiate between each tumor type using extremely high certainty. In addition, when averaged across tumor types, the macro-average AUC score was found to

be 0.9993. Hence, this score means that the model is among the best models published to date for classifying brain tumors.

4.2.3 Integration Testing

Test Case 8: End-to-End Pipeline Testing

Purpose: To validate that the integrated system (both Enhancement and Classification components) is properly functioning to assess the accuracy of each image used in the Enhancement step with "new" images.

We developed a test set which contained a total of 50 images that were never previously utilized in developing either component of the system. The test images were selected to represent numerous MRI machines, varying qualities of images, and numerous image file types (JPEG, PNG, and DICOM).

Results:

- Complete and accurate Pre-processing of all 50 test images, (100%).
- Complete and accurate Enhancement of all 50 test images, (100%).
- Complete and accurate Classification of all 50 test images, (100%).
- Average Processing Time per image was 2.8 seconds.

The integrated system was able to produce correct results for each image that was processed through the pipeline. The complete pipeline was able to accept and process complete sets of test images across multiple formats and quality levels, indicating that the system can be deployed within a real-world environment and provide effective solutions.

Test Case 9: Web Application Functionality

Assess the usability and validity of the Gradio platform with the interface.

20 images uploaded via the web interface - Predictions made by Gradio were compared against Predictions made directly from model calls for accuracy

Results:

- UI Responsiveness: Excellent (average load time = 0.3 seconds)
- Prediction Accuracy: 100% match with Direct Model Calls
- Visualization Rendering:~All charts rendered correctly
- Error Handling: Invalid Inputs were handled gracefully

The web app provides a user-friendly , intuitive experience in predicting data output while also offering Accurate Prediction results through its use of Color-codes/Digital Collages and Confidence/Safety Bars for Visual Design.

4.2.4 Edge Cases and Failure Modes

Test Case 10: Robustness to Input Variations

Identify edge cases and potential failure modes where model performance will degrade. Intentionally generated difficult data (e.g., very noisy image inputs, rotated image scans, images that exhibit a unique range of pixel intensities).

Analysis of Results:

- For completely corrupted images model predicted with a level of confidence of approximately 30%-60% rather than 90%+ as expected.
- For 45 degree rotation of an image scan model predicted accuracy was 91.2%
- Model predicted accuracy was 88.5% for an image at a low-resolution.

The model did not exhibit catastrophic failure on challenging inputs, but did exhibit reasonable degradation as anticipated due to challenging inputs. The model appropriately reflected the uncertainty associated with these challenging inputs through the assignment of confidence scores; this is ultimately beneficial in the development of clinical decision support systems. The extreme rotations and very low-resolution remain limitations for use in clinical practice.

4.2.5 Summary of Testing Outcomes

The comprehensive testing we performed produced the following conclusions:

- The improvement in the accuracy of the classifiers by adding U-Net Preprocessing shows support for using the two stages as means of classifying data.
- The EfficientNetB4 Architecture, which achieved an accuracy of 98.17% and an area under the ROC curve of 99.93%, was the highest performing architecture out of the five architectures that we tested.
- The EfficientNetB4 model maintained a constant level of performance and did not display significant weakness for any of the four Tumor Types we tested.
- Tumors from all four Tumor Classes we tested all exhibited a specificity greater than 98%, which resulted in a lower number of False Positive results.
- The Combined Input Pipeline was able to properly identify all of the tumors examined despite the differences in quality, formats, and many other different environmental input conditions found in the real world.
- As mentioned previously, the Combined Input Pipeline was limited by the decreased ability to properly process low-quality images that were in extremely high distortion levels or that were extremely low-resolution levels, which will be reviewed further as we continue to enhance our models using this methodology.